TP3: CI/CD with GitHub Actions and Render :

Étape_0 :Vérifier les prérequis sur Windows

```
C:\Users\arijb\Desktop\cicd-lab>node --version
v24.22.0

C:\Users\arijb\Desktop\cicd-lab>
```

Étape_1 : Cloner et préparer l'application de base

1. Clone le projet de base

```
C:\Users\arijb\Desktop>git clone https://github.com/wahid007/cicd-lab-starter.git cicd-lab
Cloning into 'cicd-lab'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 16.96 KiB | 1.21 MiB/s, done.

C:\Users\arijb\Desktop>cd cicd-lab
```

2.Supprime l'historique Git d'origine et initialiser

```
C:\Users\arijb\Desktop\cicd-lab>rmdir /s /q .git

C:\Users\arijb\Desktop\cicd-lab>git init -b main
Initialized empty Git repository in C:/Users/arijb/Desktop/cicd-lab/.git/

C:\Users\arijb\Desktop\cicd-lab>
```

3. Installer les dépendances :

```
C:\Users\arijb\Desktop\cicd-lab>npm ci
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.
npm warn EBADENGINE Unsupported engine {
npm warn EBADENGINE   package: 'cicd-lab@1.0.0',
npm warn EBADENGINE   required: { node: '24.x' },
npm warn EBADENGINE   current: { node: 'v22.14.0', npm: '11.8.0' }
npm warn EBADENGINE }

added 84 packages, and audited 85 packages in 18s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

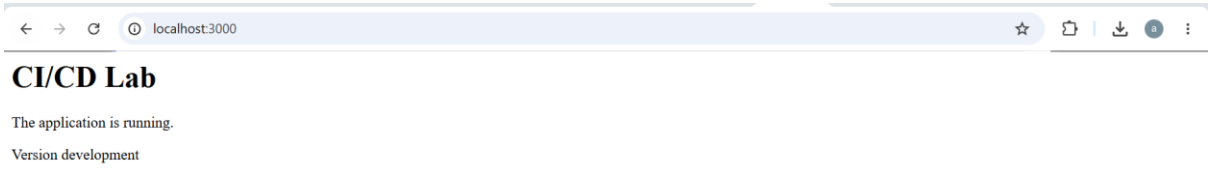
4.Lancer l'application pour tester :

```
C:\Users\arijb\Desktop\cicd-lab>npm run dev
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.

> cicd-lab@1.0.0 dev
> tsx watch src/server.ts

CI/CD Lab running at http://localhost:3000
```

5.Accéder a l'interface :



Etape2_Ajouter la commande de vérification :

1. Ajout la commande de vérification

Fichier Modifier Affichage

```
{
  "name": "cicd-lab",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "engines": {
    "node": "24.x"
  },
  "scripts": {
    "dev": "tsx watch src/server.ts",
    "typecheck": "tsc --noEmit",
    "test": "node --import tsx --test test/app.test.ts",
    "build": "tsc",
    "start": "node dist/server.js"
    "verify": "npm run typecheck && npm test && npm run build"

  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "module",
  "dependencies": {
    "express": "^5.2.1"
  },
  "devDependencies": {
    "@types/express": "^5.0.6",
    "@types/node": "^24.13.6",
    "tsx": "^4.23.15",
    "typescript": "^7.0.2"
  }
}
```

2.vérification :

```
C:\Users\arijb\Desktop\cicd-lab>npm run verify
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.

> cicd-lab@1.0.0 verify
> npm run typecheck && npm test && npm run build

npm warn Unknown env config "communication interne" (clusterip :communication interne). This will stop working in the next major vers
ion of npm.
npm warn Unknown env config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih ima
ge bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown env config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown env config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown env config "yelzem adrr ip public" (loadbalancer :yelzem adrr ip public). This will stop working in the next major v
ersion of npm.
npm warn Unknown env config "communicqtion externe nejem nehki maah mn bara" (nodeport:communicqtion externe nejem nehki maah mn bara
). This will stop working in the next major version of npm.
npm warn Unknown env config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown env config "service". This will stop working in the next major version of npm.
```

Etape_3 :Pousser le code sur GitHub

```
C:\Users\arijb\Desktop\cicd-lab>git add .

C:\Users\arijb\Desktop\cicd-lab>git commit -m "first commit"
[main (root-commit) 240fdf2] first commit
 8 files changed, 2090 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 src/app.ts
 create mode 100644 src/server.ts
 create mode 100644 test/app.test.ts
 create mode 100644 tsconfig.json

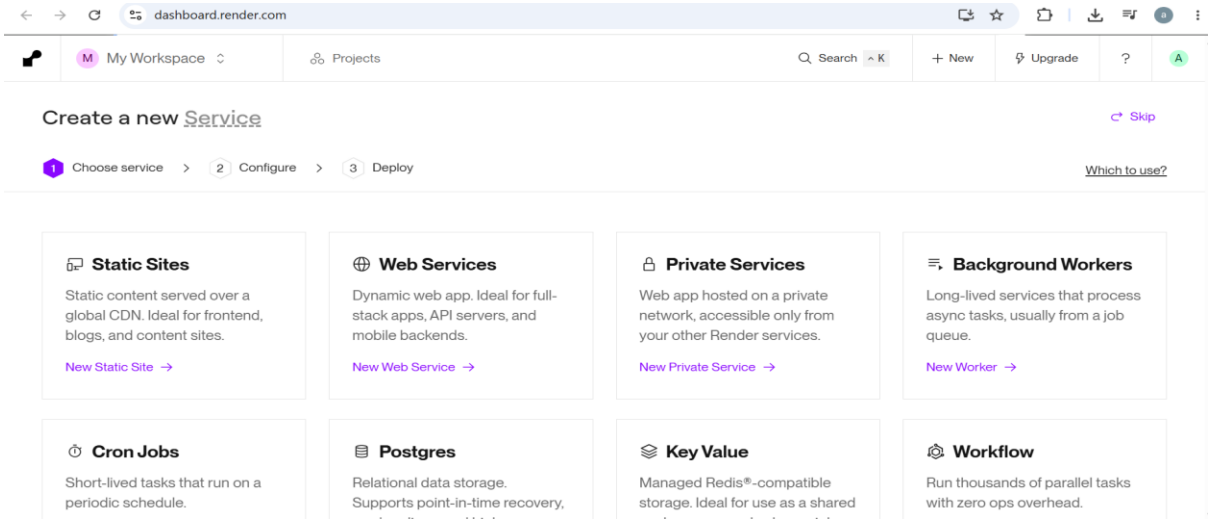
C:\Users\arijb\Desktop\cicd-lab>git remote add origin https://github.com/arijbelaid/tp3_ci-cd.git

C:\Users\arijb\Desktop\cicd-lab>git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 16.34 KiB | 3.27 MiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/arijbelaid/tp3_ci-cd.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

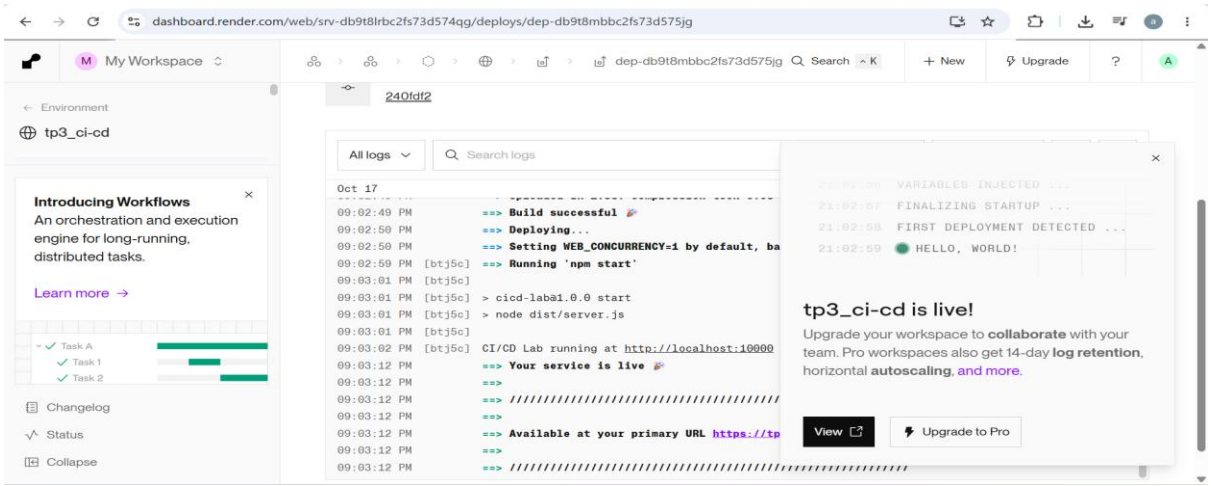
C:\Users\arijb\Desktop\cicd-lab>
```

Étape_4 : Déployer sur Render

1.accéder a Render :



2.Creer un nouveau service :



3.accees au navigateur :



Étape 5_Mettre en place l'Intégration Continue :

1. création d'une nouvelle branche

```
C:\Users\arijb\Desktop\cicd-lab>git switch -c feature/github-actions-ci
Switched to a new branch 'feature/github-actions-ci'
```

2. Création du dossier et le fichier de configuration

```
C:\Users\arijb\Desktop\cicd-lab>notepad .github/workflows/ci.yml
```

Fichier Modifier Affichage

```yaml
name: CI

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

permissions:
  contents: read

jobs:
  verify:
    name: Verify
    runs-on: ubuntu-latest
    timeout-minutes: 10

    steps:
      - name: Check out repository
        uses: actions/checkout@v4

      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 24
          cache: npm

      - name: Install dependencies
        run: npm ci

      - name: Verify application
        run: npm run verify
```

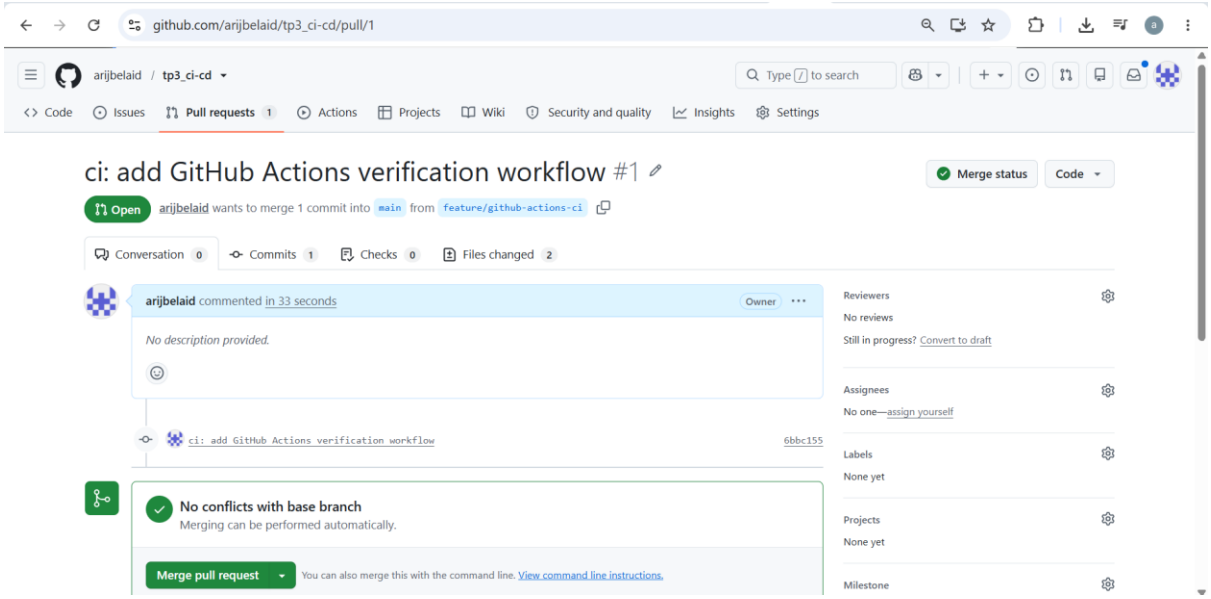
3. Pousser la branche

```
C:\Users\arijb\Desktop\cicd-lab>git add .

C:\Users\arijb\Desktop\cicd-lab>git commit -m "ci: add GitHub Actions verification workflow"
[feature/github-actions-ci 6bbc155] ci: add GitHub Actions verification workflow
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 .github/workflows/ci.yml.txt

C:\Users\arijb\Desktop\cicd-lab>git push -u origin feature/github-actions-ci
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 391 bytes | 391.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/github-actions-ci' on GitHub by visiting:
remote:      https://github.com/arijbelaid/tp3_ci-cd/pull/new/feature/github-actions-ci
remote:
To https://github.com/arijbelaid/tp3_ci-cd.git
 * [new branch]      feature/github-actions-ci -> feature/github-actions-ci
branch 'feature/github-actions-ci' set up to track 'origin/feature/github-actions-ci'.
```

4.Merger la Pull Request



5. Mettre à jour la branche:

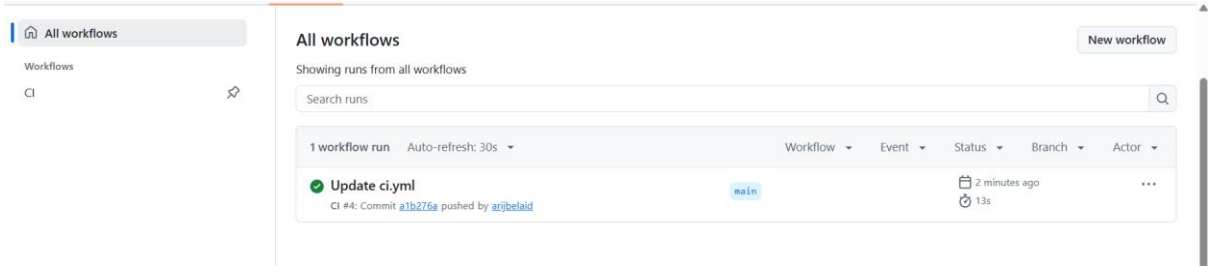
```
C:\Users\arijb\Desktop\cicd-lab>git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\arijb\Desktop\cicd-lab>git pull --ff-only
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 928 bytes | 185.00 KiB/s, done.
From https://github.com/arijbelaid/tp3_ci-cd
   240fdf2..c020cc1  main       -> origin/main
Updating 240fdf2..c020cc1
Fast-forward
 .github/workflows/ci.yml     | 0
 .github/workflows/ci.yml.txt | 0
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 .github/workflows/ci.yml.txt

C:\Users\arijb\Desktop\cicd-lab>git branch -d feature/github-actions-ci
Deleted branch feature/github-actions-ci (was 6bbc155).

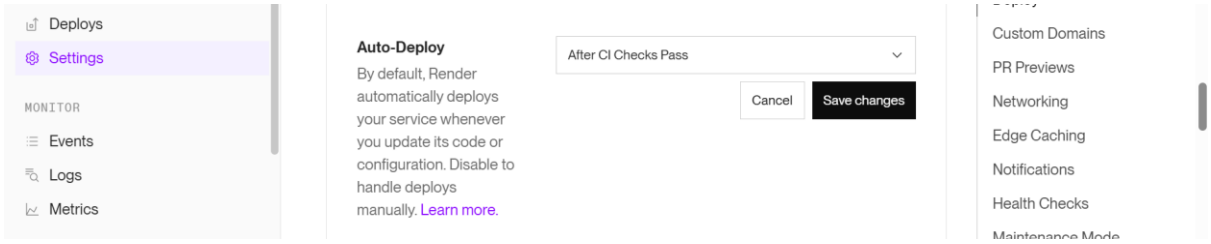
C:\Users\arijb\Desktop\cicd-lab>
```

6.Vérifier l'action :

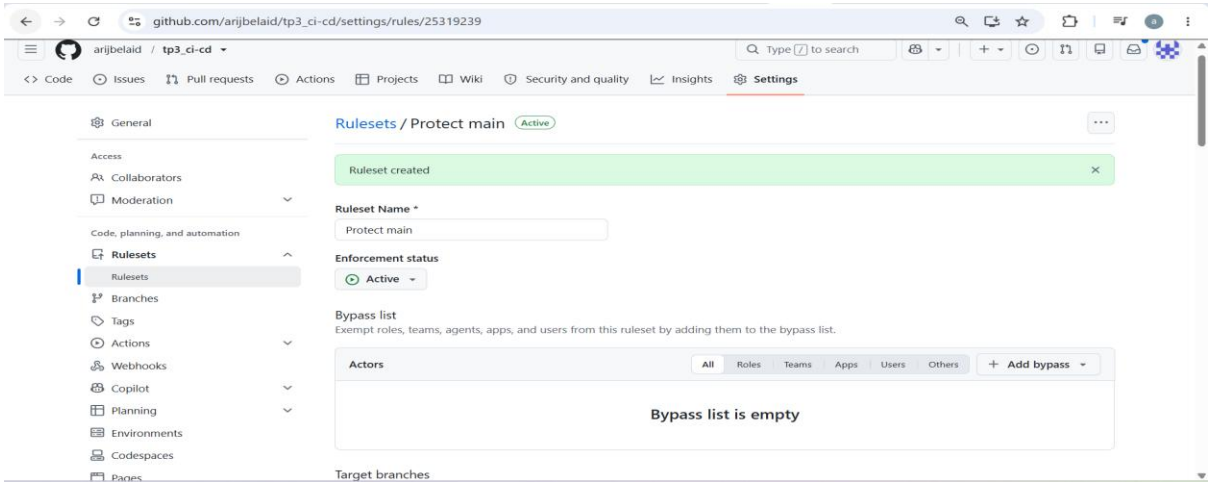


Étape_6 : Sécuriser la branche main

1.Configurer Render (Déploiement après succès des tests)



2. Configurer GitHub (Protection de la branche) :



Étape_7 : Faire un changement normal (Le flux de travail) :

1.Créer une branche :

```
C:\Users\arijb\Desktop\cicd-lab>git switch -c feature/update-homepage
Switched to a new branch 'feature/update-homepage'
```

2.modifier en src/app.ts

```
C:\Users\arijb\Desktop\cicd-lab>notepad src/app.ts
```

```
import express from 'express';

const app = express();

app.get('/', (_request, response) => {
  const version = process.env.APP_VERSION ?? 'development';

  response.type('html').send(`
    <h1>CI/CD Lab</h1>
    <p>Verified by GitHub Actions. Deployed on Render.</p>
    <p>The application is running.</p>
    <p>Version ${version}</p>
  `);
});

app.get('/api/health', (_request, response) => {
  response.json({
    status: 'ok',
  });
});

export default app;
```

3. Vérifie en local

```
C:\Users\arijb\Desktop\cicd-lab>npm run verify
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.

> cicd-lab@1.0.0 verify
> npm run typecheck && npm test && npm run build

npm warn Unknown env config "communication interne" (clusterip :communication interne). This will stop working in the next major vers
ion of npm.
npm warn Unknown env config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih ima
ge bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown env config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown env config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown env config "yelzem adrr ip public" (loadbalancer :yelzem adrr ip public). This will stop working in the next major v
ersion of npm.
npm warn Unknown env config "communicqtion externe nejem nehki maah mn bara" (nodeport:communicqtion externe nejem nehki maah mn bara
). This will stop working in the next major version of npm.
npm warn Unknown env config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown env config "service". This will stop working in the next major version of npm.
npm warn Unknown env config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next major
 version of npm.
```

4. Envoyer sur GitHub :

```
C:\Users\arijb\Desktop\cicd-lab>git add .

C:\Users\arijb\Desktop\cicd-lab>git commit -m "feat: update homepage deployment message"
[feature/update-homepage 2e926fb] feat: update homepage deployment message
 1 file changed, 1 insertion(+)

C:\Users\arijb\Desktop\cicd-lab>git push -u origin feature/update-homepage
Enumerating objects: 17, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 1.68 KiB | 245.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/update-homepage' on GitHub by visiting:
remote:      https://github.com/arijbelaid/tp3_ci-cd/pull/new/feature/update-homepage
remote:
To https://github.com/arijbelaid/tp3_ci-cd.git
 * [new branch]      feature/update-homepage -> feature/update-homepage
branch 'feature/update-homepage' set up to track 'origin/feature/update-homepage'.
```

5. mettre à jour ta branche locale :

```
C:\Users\arijb\Desktop\cicd-lab>git pull --ff-only
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (7/7), 2.18 KiB | 21.00 KiB/s, done.
From https://github.com/arijbelaid/tp3_ci-cd
   9b85896..a97ab6e  main       -> origin/main
Updating 9b85896..a97ab6e
Fast-forward
 .github/workflows/ci.yml | 34 ++++++++++++++++++++++++++++++++++
 src/app.ts               |  1 +
 2 files changed, 35 insertions(+)

C:\Users\arijb\Desktop\cicd-lab>git branch -d feature/update-homepage
Deleted branch feature/update-homepage (was 2e926fb).
```

Étape 8 _Démontrer un échec de test :

1.Crée la branche

```
C:\Users\arijb\Desktop\cicd-lab>git switch -c feature/version-endpoint
Switched to a new branch 'feature/version-endpoint'
```

2.modifier en test/app.test.ts

```
...
test('GET /api/version returns the current version', async () => {
  const response = await fetch(`${baseUrl}/api/version`);

  assert.equal(response.status, 200);
  assert.deepEqual(await response.json(), {
    version: process.env.APP_VERSION ?? 'development',
  });
});
```

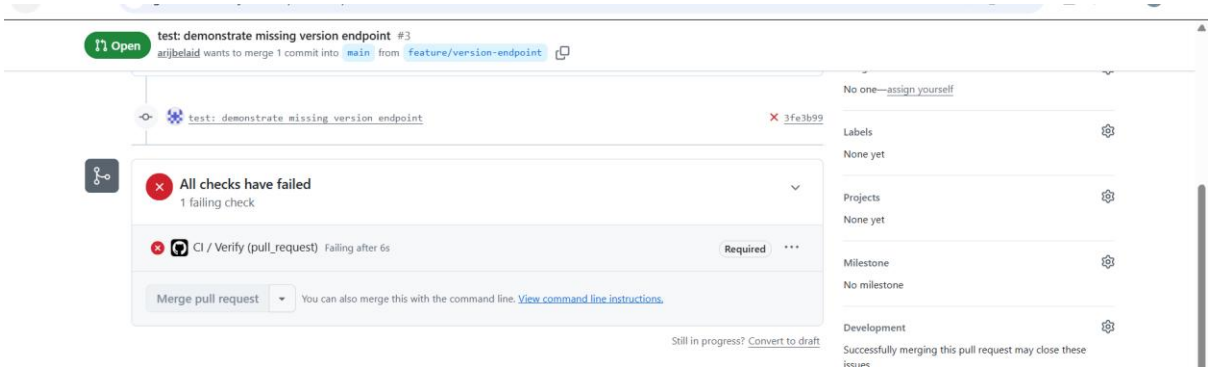
3. Envoyer le code sur GitHub

```
C:\Users\arijb\Desktop\cicd-lab>git add .

C:\Users\arijb\Desktop\cicd-lab>git commit -m "test: demonstrate missing version endpoint"
[feature/version-endpoint 3fe3b99] test: demonstrate missing version endpoint
 1 file changed, 8 insertions(+)

C:\Users\arijb\Desktop\cicd-lab>git push -u origin feature/version-endpoint
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 460 bytes | 153.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/version-endpoint' on GitHub by visiting:
remote:      https://github.com/arijbelaid/tp3_ci-cd/pull/new/feature/version-endpoint
remote:
To https://github.com/arijbelaid/tp3_ci-cd.git
 * [new branch]      feature/version-endpoint -> feature/version-endpoint
branch 'feature/version-endpoint' set up to track 'origin/feature/version-endpoint'.
```

4.Vérification :



5.Corriger l'erreur :

```
test('GET /api/health returns a healthy response', async () => {
  const response = await fetch(`${baseUrl}/api/health`);

  assert.equal(response.status, 200);
  assert.deepEqual(await response.json(), {
    status: 'ok',
  });
});

app.get('/api/version', (_request, response) => {
  response.json({
    version: process.env.APP_VERSION ?? 'development',
  });
});

export default app;
test('GET /api/version returns the current version', async () => {
  const response = await fetch(`${baseUrl}/api/version`);

  assert.equal(response.status, 200);
  assert.deepEqual(await response.json(), {
    version: process.env.APP_VERSION ?? 'development',
  });
});
```

6.Vérifier :

```
C:\Users\arijb\Desktop\cicd-lab>npm run verify
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.

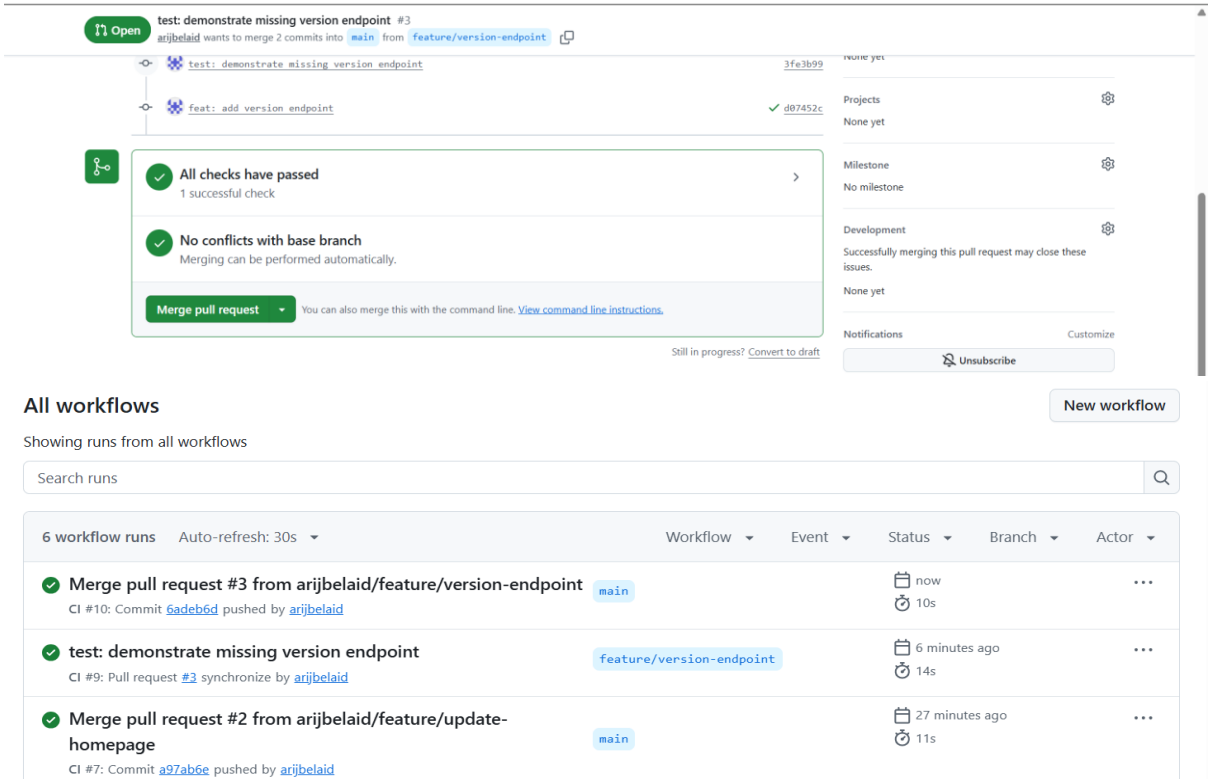
> cicd-lab@1.0.0 verify
> npm run typecheck && npm test && npm run build

npm warn Unknown env config "communication interne" (clusterip :communication interne). This will stop working in the next major vers
ion of npm.
npm warn Unknown env config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih ima
ge bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown env config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown env config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown env config "yelzem adrr ip public" (loadbalancer :yelzem adrr ip public). This will stop working in the next major v
ersion of npm.
npm warn Unknown env config "communicqtion externe nejem nehki maah mn bara" (nodeport:communicqtion externe nejem nehki maah mn bara
). This will stop working in the next major version of npm.
npm warn Unknown env config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown env config "service". This will stop working in the next major version of npm.
npm warn Unknown env config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next major
```

7. Envoyer la correction sur GitHub :

```
C:\Users\arijb\Desktop\cicd-lab>git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 430 bytes | 215.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/arijbelaid/tp3_ci-cd.git
   3fe3b99..d07452c  feature/version-endpoint -> feature/version-endpoint
```

8.Vérifier



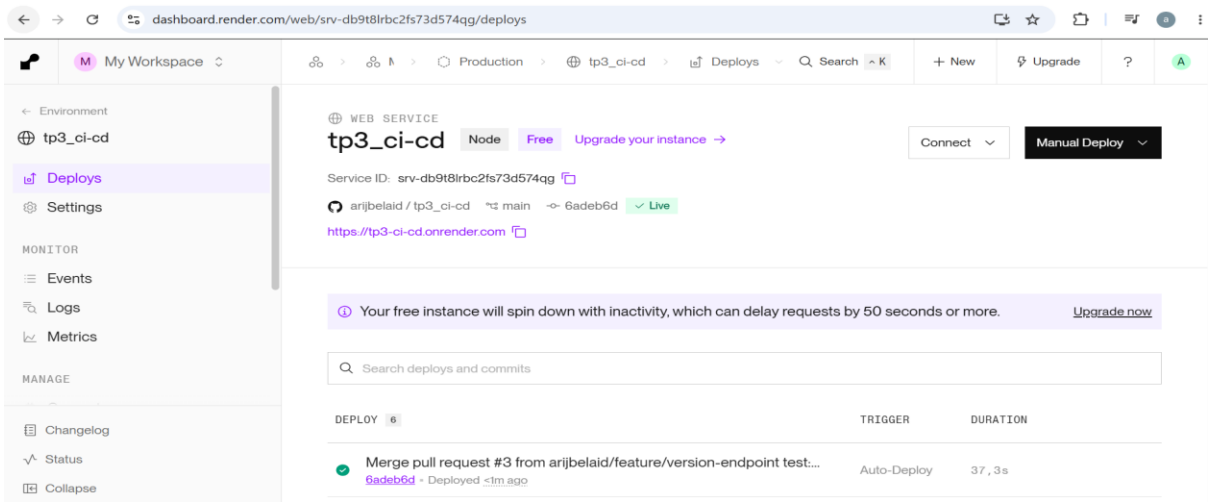
9.mettre à jour :

```
C:\Users\arijb\Desktop\cicd-lab>git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

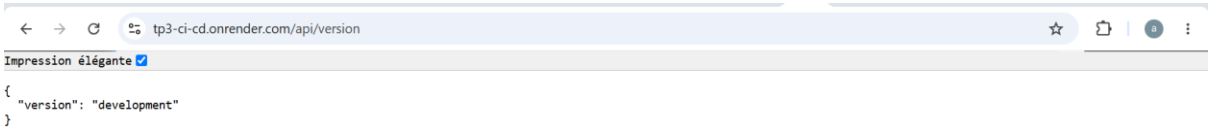
C:\Users\arijb\Desktop\cicd-lab>git pull --ff-only
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 923 bytes | 65.00 KiB/s, done.
From https://github.com/arijbelaid/tp3_ci-cd
   a97ab6e..6adeb6d  main       -> origin/main
Updating a97ab6e..6adeb6d
Fast-forward
 test/app.test.ts | 16 ++++++++++++++++
 1 file changed, 16 insertions(+)

C:\Users\arijb\Desktop\cicd-lab>git branch -d feature/version-endpoint
Deleted branch feature/version-endpoint (was d07452c).
```

10. Vérification finale sur Render

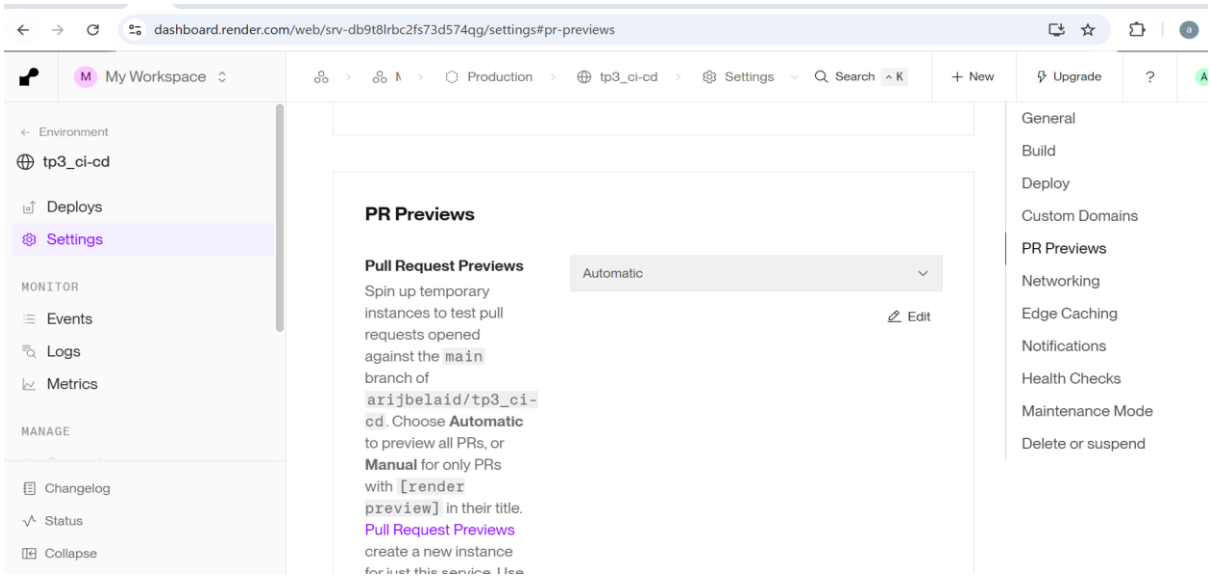


11. Vérifier l'API



Étape_9 : Les Previews :

1.mettre Previews active



2. Créer une nouvelle branche

```
C:\Users\arijb\Desktop\cicd-lab>git switch -c feature/preview-final
Switched to a new branch 'feature/preview-final'
```

3. Ajouter la nouvelle ligne

```
app.get('/', (_request, response) => {
  response.type('html').send(`
    <h1>CI/CD Lab</h1>
    <p>Test Preview Final</p>
    <p>Verified by GitHub Actions. Deployed on Render.</p>
    <p>The application is running.</p>

    <p>Version ${version}</p>
  `);
});
```

4.Vérifier

```
C:\Users\arijb\Desktop\cicd-lab>npm run verify
npm warn Unknown user config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown user config "verification du cluster(ensble ds machines connecté entre eux". This will stop working in the next majo
r version of npm.
npm warn Unknown user config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown user config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih im
age bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown user config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown user config "service". This will stop working in the next major version of npm.
npm warn Unknown user config "communicqtion externe nejem nehki maah mn bara" (NodePort:communicqtion externe nejem nehki maah mn bar
a). This will stop working in the next major version of npm.
npm warn Unknown user config "communication interne" (ClusterIP :communication interne). This will stop working in the next major ver
sion of npm.
npm warn Unknown user config "yelzem adrr ip public" (LoadBalancer :yelzem adrr ip public). This will stop working in the next major
version of npm.

> cicd-lab@1.0.0 verify
> npm run typecheck && npm test && npm run build

npm warn Unknown env config "communication interne" (clusterip :communication interne). This will stop working in the next major vers
ion of npm.
npm warn Unknown env config "objet bsh ntala3 bih app fih image bsh ntla3 nbr de pods" (deployement :objet bsh ntala3 bih app fih ima
ge bsh ntla3 nbr de pods). This will stop working in the next major version of npm.
npm warn Unknown env config "" (foundation:). This will stop working in the next major version of npm.
npm warn Unknown env config "installation" (linux :installation). This will stop working in the next major version of npm.
npm warn Unknown env config "yelzem adrr ip public" (loadbalancer :yelzem adrr ip public). This will stop working in the next major v
ersion of npm.
npm warn Unknown env config "communicqtion externe nejem nehki maah mn bara" (nodeport:communicqtion externe nejem nehki maah mn bara
). This will stop working in the next major version of npm.
npm warn Unknown env config "replicaset mas2oula aale k tih pods". This will stop working in the next major version of npm.
npm warn Unknown env config "service". This will stop working in the next major version of npm.
```

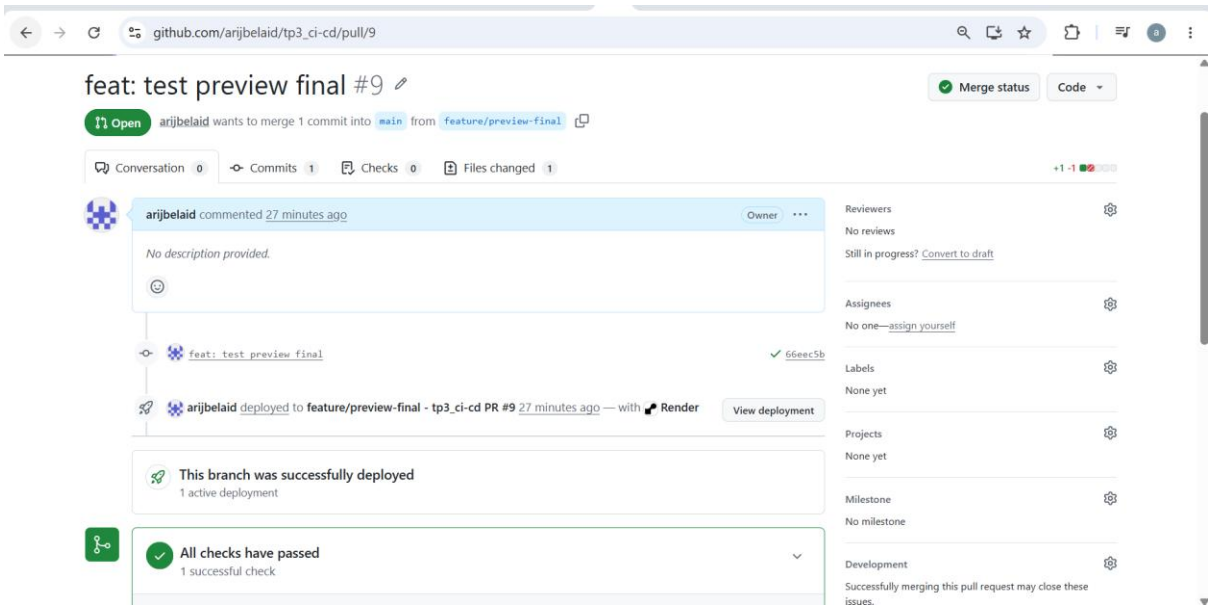
5.Envoyer sur git :

```
C:\Users\arijb\Desktop\cicd-lab>git add .

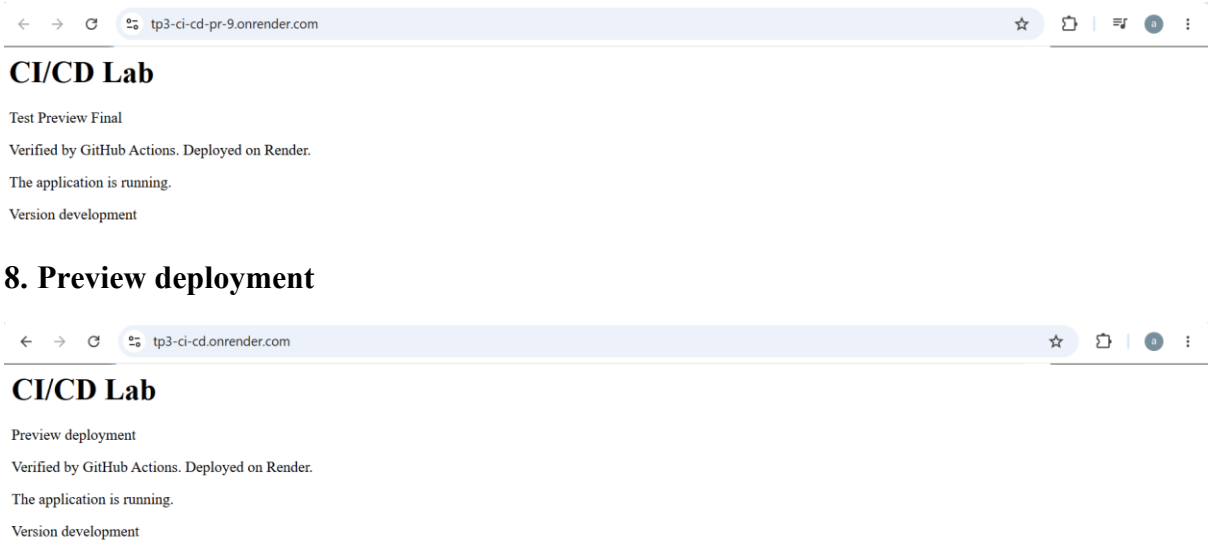
C:\Users\arijb\Desktop\cicd-lab>git commit -m "feat: test preview final"
[feature/preview-final 66eec5b] feat: test preview final
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\arijb\Desktop\cicd-lab>git push -u origin feature/preview-final
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 392 bytes | 392.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/preview-final' on GitHub by visiting:
remote:      https://github.com/arijbelaid/tp3_ci-cd/pull/new/feature/preview-final
remote:
To https://github.com/arijbelaid/tp3_ci-cd.git
 * [new branch]      feature/preview-final -> feature/preview-final
```

6. Trcrée la Pull Request



7.Test Preview Final :



8. Preview deployment